

DES - Data Encryption Standard

Block size	64 bits
Key Size	56 bits
Speed	~ 50 cycles per byte
Algorithm	Feistel Network

Timeline

- **1972** NBS call for proposals
- **1974** IBM Lucifer proposal
analyzed by DOD and enhanced by NSA
- **1976** adopted as standard
- **2004** NIST withdraws the standard

(FYI) Feistel Network

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F_i(R_{i-1}, k_i)$$

Properties:

- F is an arbitrary function that scrambles the input based on a key
 - F is not necessary invertible
 - A Feistel Network is invertible
- ➡ Achieves confusion and diffusion

